Physiosim Sprint 1

1. User Registration & Validation

+ Prerequisites:

- The internet is enabled.
- The user is unregistered (New Email).

+ Acceptance Criteria:

- **As a Student User**, I want to register a new account so that I can access the platform.
- The following registration form will be displayed:

Field	Data Type	Mandatory	Validation Rules
Full Name	String	Yes	- English Characters Only (A-Z, a-z). - Max Length: 20 Characters. - No Special Characters or Symbols.
Email	String	Yes	- Must be a valid format (user@domain.tld). - **Host Validation:** Domain must be real (e.g., gmail.com). - **TLD Validation:** Top-Level Domain must be valid (.com, .net, etc.).
Phone Number	Numeric	Yes	- Digits Only. - No special characters allowed.
Password	String	Yes	- Min 8 characters (Standard security policy).
Profile Image	Image	No	- Supported formats: JPG, PNG.

- **As a System**, upon successful registration, the user is automatically assigned the **"Basic Plan"** (Free Tier).
- **As a System**, the user's **Current IP Address** is captured and stored as the "Primary Device IP".

+ Expected Results:

- User account is created in the database.
- User is redirected to the "Home/Dashboard" page.
- The current device (PC/Laptop) is locked as the primary access point via IP.

+ Dependencies:

- Backend API for IP Capture.
- Email Validation Service.
- Email Validation Service.

---

2. Security: Device Connection (Multi-Device Locking)

+ Prerequisites:

- **User is logged in on the Primary Device (PC/Laptop via IP).**
- **User wants to connect a Mobile Device.**

+ Acceptance Criteria:

- **As a Student User, I can navigate to "Profile Settings" > "Device Connection".**
- **As a System, I will display a QR Code on the screen specifically generated for this user session.**
- **As a Mobile User, I can open the Physiosim Mobile View/App and scan the QR Code.**
- **As a System, upon scanning:**
    - **Validate that no other mobile device is currently linked to this account.**
    - **Link the Mobile Device's unique ID to the user account.**
- **As a System, if a user attempts to login from a 3rd device (Unknown IP and not the linked Mobile), access is Denied with an error message: *"Account activity restricted to registered devices only."***

+ Expected Results:

- **The account is securely bound to 1 PC (IP) and 1 Mobile (ID).**
- **Unauthorized login attempts are blocked.**

---

3. Subscription & Payment UI (Display Only)

+ Prerequisites:

- User is logged in.

+ Acceptance Criteria:

- **As a Student User**, I can view the "Subscription Plans" page.
- Three cards will be displayed:
    1. **Basic Plan:** "Free" - Access to limited Admin-selected cases only.
    2. **Semi-Annual Plan:** "1400 EGP" - Full Access for 6 Months.
    3. **Annual Plan:** "2400 EGP" - Full Access for 1 Year.
- **As a User**, when I click "Subscribe" on Paid plans, a UI placeholder for Payment Methods (Visa/Mastercard) is displayed.
- **Note:** No actual bank integration occurs in this Sprint. The button may show a "Contact Support" or "Coming Soon" message or simulate a success for testing purposes.

+ Expected Results:

- User clearly understands the pricing tiers.
- UI is ready for Payment Gateway integration in future Sprints.

---

4. Admin Panel: Case Management (CMS)

+ Prerequisites:

- User is logged in as an **Admin**.

+ Acceptance Criteria:

- **As an Admin**, I can Create, Edit, or Delete a Clinical Case.
- **Case Basic Info (The Card):**
  - **Title:** String (Mandatory).
  - **Category:** Dropdown (Ortho / Neuro / Pediatrics) (Mandatory).
  - **Difficulty:** Dropdown (Beginner / Intermediate / Advanced) (Mandatory).
  - **Is Basic?**: Checkbox (If checked, this case is visible to Free users).
  - **Cover Image:** Upload.
- **Case Internal Structure (The Logic):**
  - **Complaint:** Text Area (Mandatory - The starting point).
  - **Dynamic Steps:** Admin can add steps in any order:
    - *Step Types:* History, X-Rays, Investigations.
    - *Content:* Text description + Optional Image upload (e.g., X-ray image).
- **Question Injection:**
  - At any step, Admin can insert a Question.
  - **Type A (MCQ):** Admin inputs Question, Distractors, Correct Answer, and the "Next Step" logic.
  - **Type B (Essay/Keywords):**
    - Admin inputs the Question.
    - **Model Answer:** Text field (Hidden from student).
    - **Keywords:** Admin inputs tags (e.g., "MRI", "Spinal Cord", "Fracture").

+ Expected Results:

- A complete Case Object is saved to the database with all its steps and logic.
- The Case becomes visible in the Student Dashboard based on the "Is Basic" flag

---

5. Student Dashboard: Case Listing & Filtering

+ Prerequisites:

- User is logged in.
- Cases exist in the database.

+ Acceptance Criteria:

- **As a Student User**, I can view a grid of available Cases.
- **Card Information:** Each card displays the Case Title, Category, Difficulty Level, and Image.
- **Search & Filter:**
    - **Search Bar:** Filter by Case Title.
    - **Category Filter:** Filter by (Ortho, Neuro, Pediatrics).
    - **Difficulty Filter:** Filter by (Beginner, Intermediate, Advanced).
- **Access Control:**
    - If User is **Basic**, only cases marked "Is Basic" are clickable. Others appear locked (dimmed or with a lock icon).

+ Expected Results:

- User can easily find relevant cases to study.
- Locked content incentivizes subscription.

+ Dependencies:

- Admin must have created at least one "Basic" case and one "Paid" case for testing.

Technical Notes for Developers:

1. **Essay Grading Schema:** Ensure the database table for Essay Questions includes specific columns for Keywords (optimally stored as a JSON Array) and Model_Answer. These fields are crucial for the grading algorithm in future sprints.
2. **IP Locking Reset Mechanism:** The system must include a function for Admins to manually "Reset IP" for a user. This is a critical support requirement to handle scenarios where a student changes their device or network (Dynamic IP/Router change).
3. **Case Logic Data Structure:** The data structure for Cases must be highly flexible (e.g., using a Linked List or Graph approach). Since user answers dictate the next step, the flow is non-linear and cannot be a static sequence.